EXPERIMENT 15

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Data Preprocessing in Data Science

Aim: Handle missing values and encode categorical data.

Python Program

```
import pandas as pd
df=pd.read_csv('pre-process_datasample.csv')
df['Country']=df['Country'].fillna(df['Country'].mode()[0])
df['Age']=df['Age'].fillna(df['Age'].median())
df['Salary']=df['Salary'].fillna(df['Salary'].mean())
df=pd.get_dummies(df,columns=['Country'],dtype=int)
df['Purchased']=df['Purchased'].map({'No':0,'Yes':1})
print(df)
```

Result

Missing values are filled and categorical columns are converted into numerical form.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.